

$$\bar{M}_m = \sum_1^n x_i \bar{M}_i$$

$$\bar{H}_m = \sum_1^n \bar{H}_i = \sum_1^n n_i \bar{h}_i$$

$$\bar{h}_m = \frac{\bar{H}_m}{n_m} = \frac{\sum_1^n n_i \bar{h}_i}{n_m} = \sum_1^n x_i \bar{h}_i$$

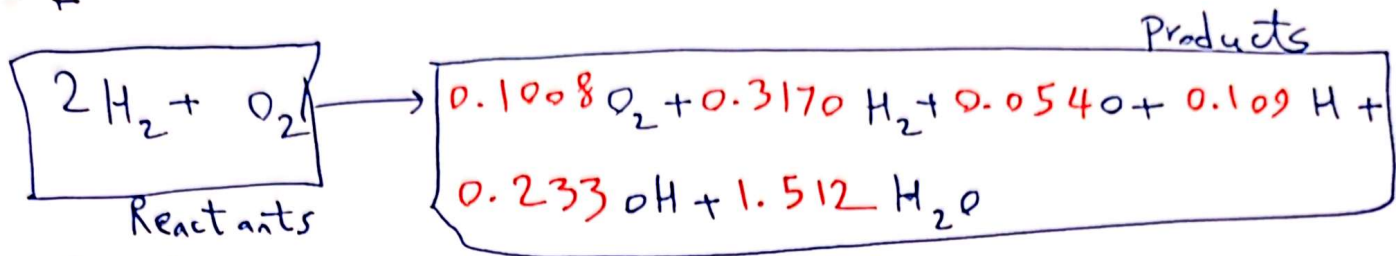
$$\bar{S}_m = \sum x_i \bar{S}_i$$

$$\bar{c}_{p,m} = \sum x_i \bar{c}_{p,i}$$

$$\gamma_m = \frac{\bar{c}_{p,m}}{\bar{c}_{v,m}} = \frac{\bar{c}_{p,m}}{\bar{c}_{p,m} - \bar{R}}, \quad R_m = \frac{\bar{R}}{\bar{M}_m}$$

Example: An ideal gas mixture consists of H_2 & O_2 and has the following properties.

Species ($\bar{M}$)	number of moles (kmol)	$\bar{c}_p$ ($\frac{kJ}{kmol \cdot K}$)	$\bar{h}$ ($\frac{kJ}{kmol}$)	x
O_2 (32)	0.1008	36.3	116233	0.0433
H_2 (2)	0.3170	32.9	105346	0.1363
O (16)	0.054	20.9	312065	0.0232
H (1)	0.109	20.9	283015	0.0469
OH (17)	0.233	33.5	149137	0.1002
H_2O (18)	1.512	48.27	-87266	0.651
	$\sum n_i = n_m$ (P-4)			



Calculate total number of moles in product, mole fraction (x), R_m , mole-specific & mass-specific enthalpies, and γ_m .

Solution: $n_m = \sum n_i = 2.3258$ (summation of all n_i)

$$X_i = \frac{n_i}{n_m} \text{ (see table)}$$

$$\bar{M}_m = \sum X_i \bar{M}_i = 15.48 \frac{\text{kg}}{\text{kmol}}$$

$$R_m = \frac{\bar{R}}{\bar{M}_m} = \frac{8314.3}{15.48} = 537 \frac{\text{J}}{\text{kg K}}$$

$$\bar{h}_m = \sum X_i \bar{h}_i = -1883 \frac{\text{kJ}}{\text{kmol}}$$

$$h_m = \frac{\bar{h}_m}{\bar{M}_m} = \frac{-1883}{15.48} = -121.7 \frac{\text{kJ}}{\text{kg}}$$

$$\bar{C}_{p,m} = \sum X_i \bar{C}_{p,i} = 42.258 \frac{\text{kJ}}{\text{kmol K}}$$

$$\gamma_m = \frac{\bar{C}_{p,m}}{\bar{C}_{p,m} - \bar{R}} = \boxed{1.24}$$

If individual $\bar{c}_{p,i}$ are not given, assume:

$$\gamma_{\text{mon}} = 1.67$$

$$\gamma_{d,i} = 1.29$$

$$\gamma_{\text{tr},i} = 1.17$$

Then get $\bar{c}_{p,i} = \frac{\gamma_i \bar{R}}{\gamma_i - 1}$, then $\bar{c}_{p,m} = \sum (X_i \bar{c}_{p,i})$

$$\gamma_m = \frac{\bar{c}_{p,m}}{\bar{c}_{p,m} - \bar{R}}$$

Ex: Reconsider the previous problem. Calculate γ_m using the γ assumption above.

$$\bar{c}_{p_o} = \bar{c}_{p_H} = \frac{\gamma_{\text{mon}} \bar{R}}{\gamma_{\text{mon}} - 1} = \frac{1.67 \bar{R}}{0.67} = 2.493 \bar{R}$$

$$\bar{c}_{p_{O_2}} = \bar{c}_{p_{H_2}} = \bar{c}_{p_{OH}} = \frac{\gamma_{d,i} \bar{R}}{\gamma_{d,i} - 1} = \frac{1.29 \bar{R}}{0.29} = 4.448 \bar{R}$$

$$\bar{c}_{p_{H_2O}} = \frac{\gamma_{\text{tr},i} \bar{R}}{\gamma_{\text{tr},i} - 1} = \frac{1.17 \bar{R}}{0.17} = 6.882 \bar{R}$$

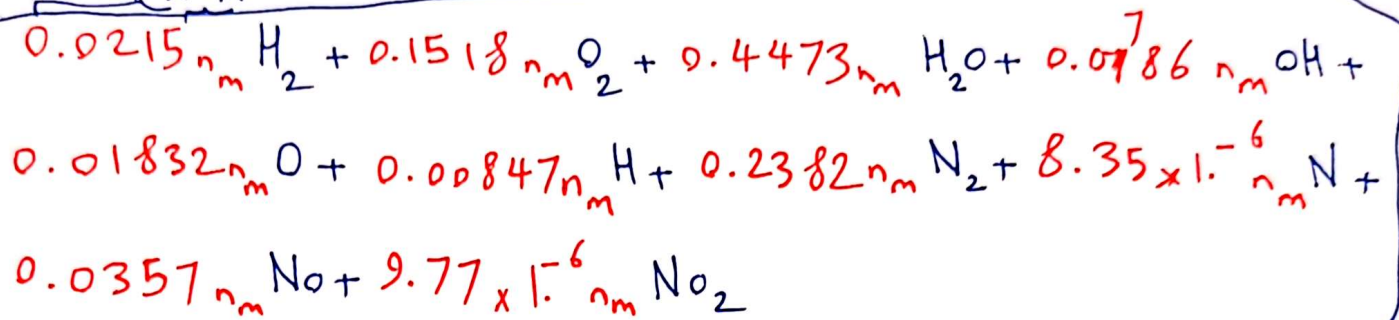
$$\bar{c}_{p,m} = \sum X_i \bar{c}_{p,i} = 5.893 \bar{R}$$

$$\gamma_m = \frac{\bar{c}_{p,m}}{\bar{c}_{p,m} - \bar{R}} = \frac{5.893 \bar{R}}{5.893 \bar{R} - \bar{R}} = 1.2$$

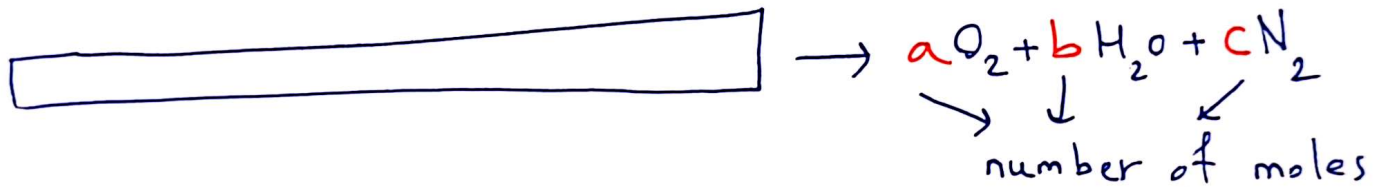
"Approximate"

Example: A mixture of gases enter a rocket nozzle with the following composition (red numbers are X_i & $n_i = X_i n_m$, n_m is the total number of moles in reactants)

$$n_i = X_i \times n_m$$



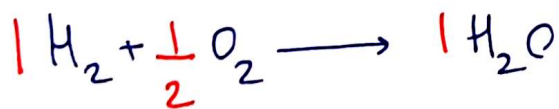
In the nozzle, recombination occurs, and we have:



Determine γ_m , $\overline{c_p}_m$ & $\overline{R}_m$ at the nozzle exit.

use $\gamma_{\text{mon}} = 1.67$, $\gamma_{\text{di}} = 1.29$, $\gamma_{\text{tr}} = 1.17$

Solution: Atoms are conserved in chemical reactions, i.e.,



$\square \times \square = \text{cte on both sides of reaction!}$

$$\left\{ \begin{array}{l} \text{for H}_2 \quad \boxed{1} \times \boxed{2} = \boxed{1} \times \boxed{2} \\ \text{for O}_2 \quad \boxed{\frac{1}{2}} \times \boxed{2} = \boxed{1} \times \boxed{1} \end{array} \right.$$

(Number of moles $\times$ number of atoms) $\equiv$ Conserved!

(P-2)

$$\begin{aligned}\text{Balance of N: } 2c &= 2 \times 0.2382 n_m + 8.35 \times 10^{-6} n_m + \\ &\quad 0.0357 n_m + 9.77 \times 10^{-6} n_m \\ &= 0.51229 n_m \Rightarrow \boxed{c = 0.2561 n_m}\end{aligned}$$

$$\begin{aligned}\text{Balance of H: } 2b &= 2 \times 0.0215 n_m + 2 \times 0.4473 n_m + 0.0786 n_m \\ &\quad + 0.09847 n_m = 1.02467 n_m\end{aligned}$$

$$\boxed{b = 0.5123 n_m}$$

$$\begin{aligned}\text{Balance of O: } 2a + b &= 2 \times 0.1518 n_m + 0.4473 n_m + 0.0786 n_m \\ &\quad + 0.01832 n_m + 0.0357 n_m + 2 \times 9.77 \times 10^{-6} n_m \\ &= 0.8838 n_m\end{aligned}$$

$$\Rightarrow \boxed{a = 0.1858 n_m}$$

$$\text{Number of moles in the product: } n_2 = a + b + c = 0.9542 n_m \neq n_m$$

number of moles is not conserved!

total atoms is conserved!

Find X_{O_2} , X_{H_2O} & X_{N_2} in product:

$$X_{O_2} = \frac{n_{O_2}}{n_{m_2}} = \frac{a = 0.1858 \text{ nm}}{0.9542 \text{ nm}} = \underline{0.1947}$$

$$X_{H_2O} = \frac{n_{H_2O}}{n_{m_2}} = \frac{b = 0.5123 \text{ nm}}{0.9542 \text{ nm}} = \underline{0.5369}$$

$$X_{N_2} = \underline{\underline{1}} - (0.1947 + 0.5369) = \underline{0.2684}$$

$$\bar{M}_m)_{\text{product}} = \sum X_i \bar{M}_i = 0.1947 \times 32 + 0.5369 \times 18 + 0.2684 \times 28 = \underline{23.41}$$